

Damilola Ogunleye

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EDUCATION

- **University College London** London, UK
Master of Engineering in Robotics and Artificial Intelligence
Overall Average: 80.43% (1st Class)
Modules: Object-Oriented Programming, Deep Learning for Robotics and Artificial Intelligence
2023 - 2027
- **St Olav VGS** Stavanger, Norway
International Baccalaureate Diploma Programme
Grade: 41/45 Maths AA HL: 7 Physics HL: 7
2021 - 2023

SKILLS SUMMARY

- **Languages & Tools:** Python, PyTorch, MATLAB, ROS2, C, scikit-learn, Git, Fusion360, PyBullet, Gazebo

EXPERIENCE

- **Google DeepMind Research Ready Programme** York, UK
Robotics Intern
Jun 2026 – Aug 2026
 - Developed a ROS2 Jazzy and Gazebo Classic simulation environment to evaluate decentralised swarm robotics for autonomous nuclear waste clustering, implementing the CacheConsensus coordination algorithm.
 - Supported the integration and validation of the autonomy stack across both simulation and physical robotic platforms, ensuring consistent behaviour between simulated and real-world deployments.
 - Engineered a computer vision pipeline for nuclear waste classification and localisation across 350 image samples, applying AMLAS assurance principles to support the safety assurance argument; achieved 90% precision and a 0.022 m RMSE localisation error.
- **National Centre for Artificial Intelligence and Robotics** Abuja, Nigeria
Robotics Intern
Aug 2025 – Sep 2025
 - Designed and prototyped an autonomous agricultural robot featuring Wi-Fi-enabled monitoring, a seed-dispensing mechanism, and an automated furrow-opening system using an Arduino-based embedded controller.
 - Implemented GPS, IMU, and wheel encoder sensor fusion using a Kalman filter, tuning measurement and process noise covariance matrices to improve localisation accuracy and navigation robustness during autonomous operation.
- **Cisco (Outshift by Cisco)** London, UK
Agentic AI Engineering Intern
Apr 2025 – Jul 2025
 - Designed and deployed a multi-agent AI workflow using LangChain and LangGraph to autonomously generate, critique, and iteratively refine CAD designs, integrating automated 3D printing from natural language specifications.

PROJECTS

- **UKSEDS Olympus Rover Trials (Advanced Stream) (November 2025 - July 2026):** Perception, Navigation and Simulation Lead. Developed a ROS2-based rover autonomy stack using RGB-D/LiDAR SLAM, A* planning, and MPPI control. Performance was optimised for Jetson/Raspberry Pi deployment performing successfully during the competition.
- **Reinforcement Learning Control of Efficient Soft Exoskeletons (September 2025 - June 2026):** Optimised assistive motion in a soft exoskeleton, achieving support equivalent to a 10% reduction in muscle activation, by designing twisted string actuators (TSAs), developing a MuJoCo-based biomechanical simulation, optimising performance with PPO.
- **Multimodal AI Literacy Agent – Lead Deep Learning Researcher (September 2025 - December 2025):** Led a cross-functional team to architect a multimodal AI agent, integrating a fine-tuned LLM (RAG and dual-dataset fine-tuning) with a YOLO/CNN vision pipeline for real-time interaction using PyTorch. Benchmarked performance against state-of-the-art commercial agents.
- **Secure Real-Time Distributed Flocking on ESP32 (FreeRTOS) (September 2025 - December 2025):** Designed and evaluated a secure real-time distributed flocking system on ESP32 using FreeRTOS, implementing deterministic task scheduling and decentralised Reynolds flocking over LoRa; built an authenticated wireless protocol with replay protection and analysed jitter, CPU usage, and communication reliability under adversarial conditions.
- **UKSEDS Olympus Rover Trials (Basic Stream) – Robotics Engineer (November 2024 - July 2025):** Led the design and construction of a tracked Martian rover, integrating autonomous QR code detection (MicroPython) and a custom mobile interface for teleoperation across simulated terrain. Secured the 'Best Innovation' award and 2nd place overall.
- **A Stochastic Simulation Framework for Robotic Grasp Stability Classification. (October 2024 - December 2024):** Engineered an Object-Oriented PyBullet simulation to stochastically generate grasping datasets, extracting kinematic and contact features to train PyTorch MLPs and Scikit-learn SVMs, achieving 79% accuracy in predicting grasp stability.

HONORS AND AWARDS

- Best Innovation Award & 2nd Place Overall – UKSEDS Olympus Rover Trials (2025)
- **1st Place** – Google BGN Hackathon (Spotlight by Gemi9), Oct. 2024

EXTRACURRICULARS

- Clarinettist with the All Souls Orchestra; former member of the Rogaland Regional Youth Symphony Orchestra.
- Member of the UCL AI Society, Computer Science Society, and Jazz Society.